

sensor features a global shutter and has both monochrome and colour options in its line-up. XGS 16000

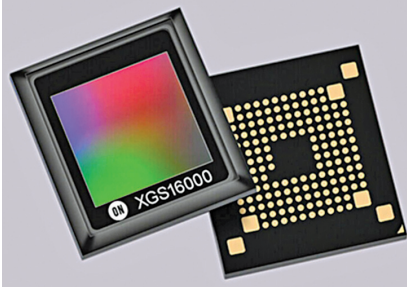


Fig. 8: On Semiconductor XGS16000



Fig. 9: Toshiba Teli DDU1607MG/MC which uses XGS 16000 (Source: Link)

sports one of the highest resolutions of 4000×4000 pixels for a standard dimension of 29×29 mm industrial camera. The use of a standard size makes this sensor compatible with many lenses and camera bodies. It uses the HiSpi output interface for communicating.

The XGS 16000 is ideal for inspection and industrial camera systems, security cameras, and other machine vision application. This sensor being used in Toshiba Teli Corporation's DDU1607MG/MC is available in an iLGA package that measures 29×29 mm. It is ideal for industrial camera systems, inspection systems (food, bottles, recycling labels, etc), security systems, and machine vision camera systems.

SonyIMX 478. Sony launched the IMX 478 in September 2021. The imaging sensor can take pictures up

to 8.13 megapixel with an astounding speed of 194fps. The 1.7cm CMOS image sensor is optimised for UV wavelengths between 200 and 400nm and has one of the smallest pixel size of $2.74 \times 2.74\mu\text{m}$ currently available in any UV sensor. It features Sony's Pregius S global shutter, which claims to provide distortion-free image and high-speed performance.

The UV wavelength sensors are widely used in semiconductor industry for pattern defect detection. It is also used for plastic sorting, scratch detection and various other applications. Fig. 10 compares images of the protection resin applied on the substrate line under visible light (left) and UV light (right).

G-Pixel GTOF0503. The GTOF0503 is a CMOS image sensor featuring the indirect Time of Flight (iTOF) technology. The sensor has a resolution of 640×480 pixels, which is around 0.41MP with pixel size of $5\mu\text{m}$. It is capable of taking good-quality pictures in challenging light conditions. It uses industry standard MIPI CSI-2 interface, making it a very versatile option. The imaging sensor can be used for 3D mapping, depth sensing, and distance measurement applications thanks to its iTOF technology. It can also be used in vision guided robots, pick and pack robots, etc. Fig. 11 shows the 3D point cloud generated from IR depth map.

On Semiconductor AR0821CS.

The AR0821CS from On Semi was released in the last quarter of 2021. The new 1.5mm CMOS sensor can capture 8.3MP pictures and features a rolling shutter and embedded High Dynamic Range (eHDR) technology. The company claims that it provides best-in-class dynamic range in challenging lightning conditions. This sensor has been used in Basler area scan sensor module daA3840-30mc shown in Fig. 12. The imaging sensor is specially designed for machine vision cameras, drones, scanners,

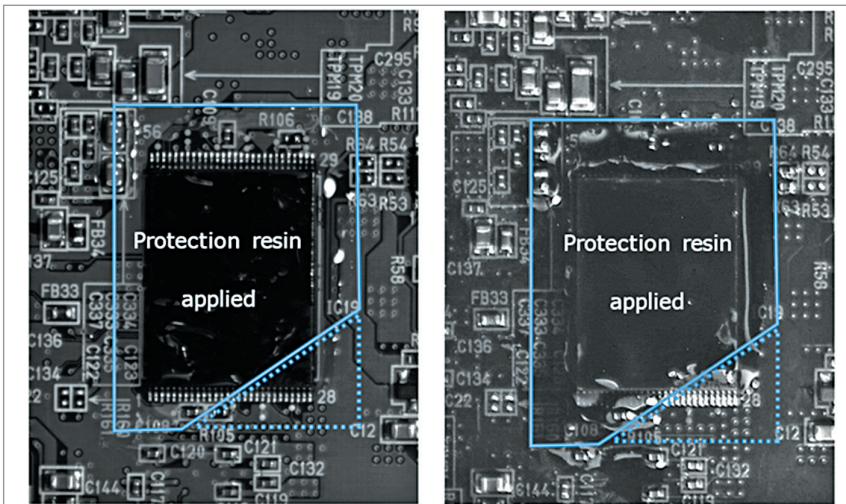


Fig. 10: Images of protection resin applied on the substrate line under visible light (left) and UV light (right). The protection resin appears black under UV lens (Source: Link)

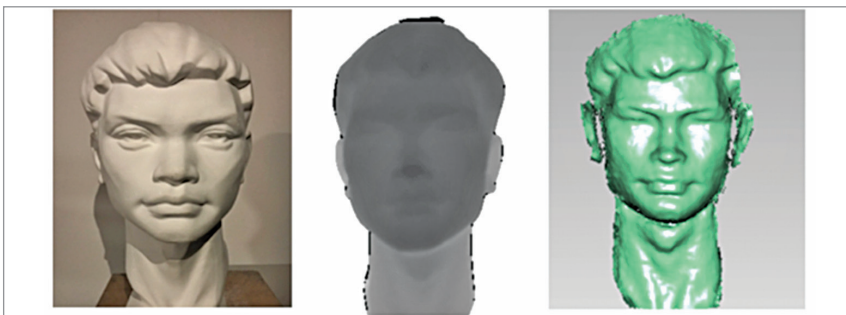


Fig. 11: The 3D point cloud generated from an IR depth map (Source: Link)